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Inventor(s)	Lai; Shu-Ting et al.

Semiconductor package

Abstract

A carrier structure is provided with a plurality of package substrates connected via connecting sections, and a functional element and a groove are formed on the connecting section, such that the groove is located between the package substrate and the functional element. Therefore, when a cladding layer covering a chip is formed on the package substrate, the groove can accommodate a glue material overflowing from the cladding layer to prevent the glue material from contaminating the functional element.

Inventors: Lai; Shu-Ting (Taichung, TW), Li; Chiu-Lien (Taichung, TW), Su; Che-Min (Taichung, TW), Hung; Chun-Huan (Taichung, TW), Hsieh; Mu-Hung (Taichung, TW), Yao; Cheng-Han (Taichung, TW), Directo; Fajanilan Darcyjo (Taichung, TW), Hsu; Cheng-Liang (Taichung, TW)

Applicant: SILICONWARE PRECISION INDUSTRIES CO., LTD. (Taichung, TW)

Family ID: 1000008821467

Assignee: SILICONWARE PRECISION INDUSTRIES CO., LTD. (Taichung, TW)

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Primary Examiner: Harriston; William A

Attorney, Agent or Firm: Studebaker Brackett PLLC

Background/Summary

BACKGROUND

1. Technical Field

(1) The present disclosure relates to a semiconductor packaging process, and more particularly, to a carrier structure that can improve the reliability of the process.

2. Description of Related Art

(2) In the early development of semiconductor packaging, a lead frame was used as a carrier for carrying active elements, the main reason being that the lead frame has the advantages of lower manufacturing cost and higher reliability. However, with the vigorous development of the electronic industry, electronic products are trending toward light, thin and short in terms of type

and high-performance, high-functionality and high-speed in terms of function. Therefore, in order to meet the requirements of high integration and miniaturization of semiconductor devices, lead frames are gradually replaced by package substrates with high-density and fine-pitch circuits in the current packaging process.

(3) As shown in FIG. 1A, in a conventional packaging process, a plurality of package substrates **10** are arrayed into a substrate strip **1**, and a plurality of positioning holes **100** are formed on the periphery of the substrate strip **1** corresponding to the corners of each of the package substrates **10**, and a plurality of test pads **101** and at least one barcode **102** are arranged on the edge of the substrate strip **1**, so that in the subsequent packaging process, as shown in FIG. 1B, semiconductor chips **30** are arranged on each of the package substrates **10**, and then the semiconductor chips **30** are covered with an encapsulant **32** to obtain a plurality of semiconductor packages **3**, and then the batch number of each of the semiconductor packages **3** is identified via the barcode **102** to read the relevant information of the package substrate **10**.

(4) The package substrate **10** includes a plurality of dielectric layers **10b** and a plurality of circuit layers **10a** disposed on the plurality of dielectric layers **10b**.

(5) The semiconductor chip **30** is electrically connected to the plurality of circuit layers **10a** via a plurality of conductive bumps **31** in a flip-chip manner, and the electrical connection between the semiconductor chip **30** and the plurality of circuit layers **10a** is tested via the test pads **101**.

(6) However, when the conventional substrate strip **1** is in the packaging process, it is difficult to control the flow of the encapsulant **32**, so that a glue material **320** of the encapsulant **32** easily overflows outside the package substrate **10** and often covers the test pads **101** or the barcodes **102**, or even the positioning holes **100**, causing the test pads **101** or the barcodes **102** to be stained, so that the test pads **101** or the barcodes **102** cannot be used subsequently.

(7) Therefore, how to overcome the above-mentioned drawbacks of the prior art has become an urgent issue to be solved at present.

SUMMARY

(8) In view of the various deficiencies of the prior art, the present disclosure provides a carrier structure, comprising: at least one package substrate; at least one connecting section adjacent to the package substrate; at least one functional element disposed on the connecting section; and at least one groove formed on the connecting section and located between the package substrate and the functional element.

(9) In the aforementioned carrier structure, the package substrate has at least one dielectric layer and at least one circuit layer formed on the dielectric layer. The present disclosure further comprises an insulating protection layer formed on the outermost dielectric layer of the package substrate.

(10) In the aforementioned carrier structure, the connecting section includes at least one insulating layer, and the functional element is formed on the insulating layer. For example, a depth of the groove is a thickness of the insulating layer.

(11) In the aforementioned carrier structure, the connecting section is formed with a positioning hole corresponding to a corner of the package substrate.

(12) In the aforementioned carrier structure, the functional element is a positioning hole, a test pad, or a barcode.

(13) In the aforementioned carrier structure, a length of the groove is greater than or equal to a maximum corresponding length of a layout area of the functional element.

(14) In the aforementioned carrier structure, a main distance is defined between a center of the functional element and an edge of the package substrate, and a target distance is defined between a center of the groove and the edge of the package substrate, such that a ratio of the target distance to the main distance is 0.3 to 0.5. Further, the ratio of the target distance to the main distance is 0.34 to 0.46.

(15) As can be seen from the above, in the carrier structure of the present disclosure, the groove is

arranged between the package substrate and the functional element, so as to accommodate the glue material overflowing from the encapsulant. Therefore, compared with the prior art, when the carrier structure of the present disclosure is in the packaging process, even if the glue material of the encapsulant overflows onto the connecting section outside the packaging substrate, the glue material will flow into the groove and will not spread to the functional element, such that the problem of contamination of the functional element can be avoided, so that the functional element can be effectively used in the future.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1A is a schematic top plan view of a conventional substrate strip.
- (2) FIG. 1B is a schematic partial cross-sectional view of a conventional semiconductor package.
- (3) FIG. 2A is a schematic top plan view of a carrier structure according to the present disclosure.
- (4) FIG. 2B is a schematic partial cross-sectional view of a semiconductor package made by the carrier structure according to the present disclosure.
- (5) FIG. 3A and FIG. 3B are schematic partially enlarged views of FIG. 2A.

DETAILED DESCRIPTIONS

- (6) The following describes the implementation of the present disclosure with examples. Those skilled in the art can easily understand other advantages and effects of the present disclosure from the contents disclosed in this specification.
- (7) It should be understood that, the structures, ratios, sizes, and the like in the accompanying figures are used for illustrative purposes to facilitate the perusal and comprehension of the contents disclosed in the present specification by one skilled in the art, rather than to limit the conditions for practicing the present disclosure. Any modification of the structures, alteration of the ratio relationships, or adjustment of the sizes without affecting the possible effects and achievable proposes should still be deemed as falling within the scope defined by the technical contents disclosed in the present specification. Meanwhile, terms such as “on,” “upper,” “one,” “a,” and the like used herein are merely used for clear explanation rather than limiting the practicable scope of the present disclosure, and thus, alterations or adjustments of the relative relationships thereof without essentially altering the technical contents should still be considered in the practicable scope of the present disclosure.
- (8) FIG. 2A is a schematic top plan view of a carrier structure 2 according to the present disclosure. As shown in FIG. 2A, the carrier structure 2 includes: a plurality of package substrates 20 arranged in an array, a plurality of connecting sections 23 connected to the plurality of package substrates 20, and a plurality of grooves 210 disposed on the plurality of connecting sections 23.
- (9) In an embodiment, the carrier structure 2 is of a full-panel type, such as a substrate strip specification, and the connecting section 23 is configured with functional elements 2a, such as positioning holes 200, test pads 201, or a barcode 202, and a plurality of the positioning holes 200 are formed on an area of the connecting section 23 corresponding to the corners of each of the package substrates 20. For example, the positioning holes 200 are used for accommodating positioning pins on the machine table, and the positioning holes 200 are circular. However, there are many kinds of shapes and arrangement positions of the positioning holes 200, which can be designed according to requirements, and the present disclosure is not limited to as such.
- (10) Furthermore, the positioning holes 200 can also be used as an alignment mechanism in the subsequent packaging process. For example, when placing an electronic element such as a semiconductor chip 30 (as shown in FIG. 2B) on the package substrate 20, the electronic element can be aligned via the positioning holes 200; alternatively, when performing a lamination process or a molding process, the upper and lower molds can be accurately positioned on the carrier

structure **2** via the positioning holes **200** to facilitate the formation of a cladding layer (such as the encapsulant **32** as shown in FIG. 2B) for covering the electronic element.

(11) In addition, the electronic elements are arranged on each of the package substrates **20** according to the required number, and the electronic elements can be active elements, passive elements, or combinations thereof, etc., wherein the active elements are, for example, semiconductor chips, and the passive elements are, for example, resistors, capacitors, and inductors. For example, the electronic element is a semiconductor chip, which can be electrically connected to the package substrate **20** by wire bonding; alternatively, the electronic element can be disposed on the package substrate **20** in a flip-chip manner via the plurality of conductive bumps **31** (as shown in FIG. 2B); or, the electronic element can be embedded in the package substrate **20**. It should be understood that there are many ways to configure the electronic element and electrically connect the package substrate **20**, and the present disclosure is not limited to as such.

(12) In addition, the cladding layer is made from an insulating material, such as polyimide (PI), dry film, epoxy resin, molding compound, or other encapsulant **32**, and the present disclosure is not limited to as such.

(13) Therefore, the carrier structure **2** can be applied to a semiconductor packaging process to obtain a plurality of semiconductor packages **3**.

(14) The package substrate **20** is a circuit structure with a core layer or a coreless circuit structure, wherein the package substrate **20** has at least one dielectric layer **20b** and at least one circuit layer **20a** (such as a fan-out type redistribution layer [RDL]) disposed on the dielectric layer **20b**, and an insulating protection layer **21** such as a solder mask layer is formed on the outermost dielectric layer **20b**.

(15) In an embodiment, the package substrate **20** is rectangular, and the main material for forming the dielectric layer **20b** is polybenzoxazole (PBO), polyimide (PI), prepreg (PP) and other dielectric materials. It should be understood that the package substrate **20** can be selected to form the insulating protection layer **21** or not to form the insulating protection layer **21** according to requirements.

(16) However, the package substrate **20** may also be other carriers for carrying chips, such as organic plates, semiconductor materials, or other carrier plates with metal routing, and the present disclosure is not limited to as such.

(17) The connecting section **23** is disposed around the edge **20c** of the package substrate **20**, and the connecting section **23** includes at least one insulating layer, so that the functional element **2a** is formed on the insulating layer.

(18) In an embodiment, the structure of the connecting section **23** can be fabricated according to the process and structure of the package substrate **20**. For example, the insulating layer can be the dielectric layer **20b** and/or the insulating protection layer **21**, without forming a circuit layer, so that the functional element **2a** and the positioning hole **200** are formed on the insulating protection layer **21**.

(19) Furthermore, the connecting section **23** may form a circuit layer **20a** at which corresponds to each positioning hole **200**, so that the positioning pin contacts the circuit layer **20a** in the positioning hole **200** for grounding and antistatic.

(20) In addition, the connecting section **23** is defined between each of the package substrates **20** as a cutting path for the singulation process.

(21) The groove **210** is formed on the insulating protection layer **21** and located between the package substrate **20** and the functional element **2a**.

(22) In an embodiment, the depth **h** of the groove **210** is the thickness **d** of a single insulating layer (such as the insulating protection layer **21**), such that the groove **210** penetrates through the single insulating layer (or as shown in FIG. 2B, the surface of the dielectric layer **20b** is exposed).

(23) Please refer to FIG. 3A and FIG. 3B together, lengths **L1** and **L3** of the groove **210** are greater than or equal to the maximum corresponding lengths **L2** and **L4** of the layout area of the functional

element **2a**. For example, the arrangement range of the groove **210** corresponds to the test pad **201**, as shown in FIG. 3A, so that the length **L1** of the groove **210** is greater than or equal to the maximum corresponding length **L2** of the test pad **201**. Alternatively, as shown in FIG. 3B, the arrangement range of the groove **210** can correspond to the test pad **201** and the barcode **202** together, so that the length **L3** of the groove **210** is greater than or equal to the maximum corresponding length **L4** of the layout area of the test pad **201** and the barcode **202**.

(24) In addition, as shown in FIG. 3A or FIG. 3B, the groove **210** is rectangular, wherein a main distance **t** is defined between the center of the functional element **2a** (such as the test pad **201** or the barcode **202**) and the edge **20c** of the package substrate **20**, a first distance **t1** is defined between the center of the groove **210** and the center of the functional element **2a**, and a second distance **t2** is defined between the center of the groove **210** and the edge **20c** of the package substrate **20**, so that the second distance **t2** is used as a target distance, and the ratio of the second distance **t2** to the main distance **t** is 0.3 to 0.5, that is, $t2/(t1+t2)=0.3$ to 0.5, preferably 0.34 to 0.46. It should be understood that the edge **20c** of the package substrate **20** is the boundary between the package substrate **20** and the connecting section **23**.

(25) Therefore, in the carrier structure **2** of the present disclosure, the groove **210** is arranged between the package substrate **20** and the functional element **2a**, so as to accommodate the glue material **320** overflowing from the encapsulant **32**. Therefore, compared with the prior art, when the carrier structure **2** of the present disclosure is in the packaging process, even if the glue material **320** of the encapsulant **32** overflows onto the connecting section **23** outside the package substrate **20**, the glue material **320** will flow into the groove **210** and will not spread to the functional element **2a**, such that the problem of contamination of the functional element **2a** can be avoided, so that the functional element **2a** can be effectively used in the future.

(26) Furthermore, the depth **h** or the lengths **L1**, **L3** of the groove **210** can match the amount of the overflowing glue material **320** in the subsequent packaging process, so that the glue material **320** cannot cover part or all of the surface of the functional element **2a**.

(27) Furthermore, the position of the groove **210** can be adjusted according to the flow range of the glue material **320**, so that the ratio of the second distance **t2** (target distance) to the main distance **t** is 0.3 to 0.5, preferably 0.34 to 0.46, so that the glue material **320** cannot cover part or all of the surface of the functional element **2a**.

(28) To sum up, the carrier structure of the present disclosure can effectively prevent the glue material of the cladding layer from contacting the functional element via the design of the groove. Therefore, in the semiconductor packaging process, the carrier structure of the present disclosure can effectively use the test pad or the barcode, so as to improve the yield and production capacity of the semiconductor packaging process.

(29) The foregoing embodiments are provided for the purpose of illustrating the principles and effects of the present disclosure, rather than limiting the present disclosure. Anyone skilled in the art can modify and alter the above embodiments without departing from the spirit and scope of the present disclosure. Therefore, the scope of protection with regard to the present disclosure should be as defined in the accompanying claims listed below.

Claims

1. A semiconductor package, comprising: at least one package substrate; an electronic element disposed on the package substrate; a cladding layer covering the electronic element; at least one connecting section adjacent to the package substrate; at least one functional element disposed on the connecting section; and at least one groove formed on the connecting section and located between the package substrate and the functional element, wherein the groove accommodates a part of a glue material of the cladding layer so that the glue material does not contact the functional element.

2. The semiconductor package of claim 1, wherein the package substrate has at least one dielectric layer and at least one circuit layer formed on the dielectric layer.
 3. The semiconductor package of claim 2, further comprising an insulating protection layer formed on the outermost dielectric layer of the package substrate.
 4. The semiconductor package of claim 1, wherein the connecting section includes at least one insulating layer, and the functional element is formed on the insulating layer.
 5. The semiconductor package of claim 4, wherein a depth of the groove is a thickness of the insulating layer.
 6. The semiconductor package of claim 1, wherein the functional element is a positioning hole, a test pad, or a barcode.
 7. The semiconductor package of claim 1, wherein a length of the groove is greater than or equal to a maximum corresponding length of a layout area of the functional element.
 8. The semiconductor package of claim 1, wherein a main distance is defined between a center of the functional element and an edge of the package substrate, and a target distance is defined between a center of the groove and the edge of the package substrate, such that a ratio of the target distance to the main distance is 0.3 to 0.5.
 9. The semiconductor package of claim 8, wherein the ratio of the target distance to the main distance is 0.34 to 0.46.
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